

Hello my name is Carlota Corbella and I am a phd student in the Climatology Group at the university of Bern. Today I am going to present a dataset that I've been working on for almost the past year, and which unifies daily and subdaily weather observations across Europe before the 20th century. This is the same dataset that I presented this May in EGU, so please bear with me those who were already there, but now as promised it's almost ready to be open to the public, so bear with me.

The main goal of this dataset, which we called HIST-DAILY, is to unify and standardize daily and subdaily observations that at the moment are scattered across archives, national services, and old publications. There's many of these: the Copernicus Climate Data Store, ECA&D, GHCNd, ISPD, etc. but none to our understanding that brings all these together.

Let's start with the motivation of building a daily dataset. Because we do have very good monthly data and well-used datasets like HCLIM. I'll go over this quickly, as many of you don't need convincing. But let me show you what monthly data alone can miss. Here's the monthly temperature of Padua through the 1830s, as published in the IMPROVE initiative.

Only once, in January 1830, did the monthly mean temperature drop below freezing.

That single cold month stands out. But at monthly resolution, that's all we see: one cold dot.

When we superimpose the daily data, we see a different story, with days that swing well above or below the monthly mean.

Daily departures from the monthly mean can reach up to 8 degrees. And this is just for temperature, for precipitation for example the effect is more dramatic, since a single flood day is nothing like continuous light rain, even if the monthly totals match.

HIST-DAILY, however, not only provides daily data, more than half of our series are subdaily. Let's take the cold winter of 1829/1830 for example.

In this graph the daily means are shown as bars, and the sub-daily observations as the line. Looking at the data, January wasn't simply a cold month, we can see what appears to be two pronounced cold spells separated by a brief recovery. The sub-daily observations reveal the within-day structure. Temperatures dropped to -16°C but the diurnal cycle was preserved. This kind of structure is invisible at monthly resolution, here represented with the blue horizontal bars, so without daily data we would never know whether the cold was sustained, episodic, or how the events were separated in time. Interesting of these couple of cold days is as well this cold dot which looks like an outlier. This is something that the Quality Control later on will detect, and which we suspect is a wrong digitization of the temperature sign.

So let me show you what's in HIST-DAILY first, and then walk through how we built the database.

We collected data from 88 sources, both already published, like the Copernicus Climate Data Store, ECA&D, GHCNd, ISPD, and not yet published from own digitizations at the university or collaborations. In total, more than 1800 series.

The focus has been on pressure and temperature, the most widely recorded variables. But we've also included non-instrumental observations such as wind direction. A key strength of HIST-DAILY is that for many sites we provide multiple variables together, which enables studies of compound events and the synoptic dynamics behind them.

This figure shows how the available network evolves over time. Keep in mind that this is what has been digitized and collected, and many more records exist in archives that haven't yet been rescued and also some that are privately kept. A few features stand out. First, the peak in the 1780s is from the Palatine Society, a coordinated European observing network with standardized instruments. Then from the mid-19th century onward, as national meteorological services began to be established, the network grows steadily.

As for the spatial extent, temperature and pressure offer the broadest coverage, from the Iberian Peninsula to Scandinavia and into Russia., Precipitation clusters in Britain and Ireland and the Alpine area, and wind direction is concentrated in central Europe. Coverage is thinner in southeastern Europe and across much of Scandinavia before the mid-19th century.

Now let me walk you through how we built the dataset. We started with two streams of input data.

The pre-processing step includes unit conversion from historical scales to the international system, time-of-day adjustment and, most crucially for the daily scale, calendar harmonization, as many sources adopted the Gregorian calendar at different points in history. We detected the calendar shifts by comparing the temperature and pressure anomalies, when available, with the corresponding cells in 20CRv3.

Afterwards, the data were submitted to various quality control steps. When QC tests flagged values, we went back to the original sources, where possible, to distinguish digitization errors from real extremes. Clear errors were corrected, and the process was repeated. Values that couldn't be verified were flagged but not deleted, thus preserving traceability.

Then, together with rigorous metadata, the final file is produced in the Station Exchange Format, or SEF.

The SEF files are a tab-separated text format, and it looks like this. A minimal header carries all station metadata, and each row corresponds to one observation with its original value and any corrections documented in the metadata column. The idea is that the dataset will be easy to integrate into reanalysis pipelines and other rescue efforts.

Now let's go for a validation example, the cold winter of 1829/30, one of the coldest winters on record in Europe, that's the one I showed at the beginning. The top panel shows daily temperature anomalies from five northern Italian stations, compared to the 20th century reanalysis ensemble mean and spread for the corresponding cells. Two things stand out. First, the 5 stations agree remarkably well with each other, which gives us confidence that our quality control preserves real climate signals. Second, the reanalysis follows the observations closely in timing but consistently underestimates the depth of the cold extremes. The bottom panels show this quantitatively, as the observed distribution has heavier cold tails than 20CR reproduces. The panel on the bottom right shows pairwise correlations between the stations, all above 0.8, confirming internal consistency across independent records.

This case study shows that HIST-DAILY can capture extremes that current reanalyses underestimate, with the kind of information needed to improve them.

To summarize: HIST-DAILY brings together more than 1,700 daily and sub-daily series from across Europe before 1900 standardized in SEF format. It combines multiple variables at the same locations, enabling studies of past extreme events and synoptic dynamics that monthly data cannot resolve. Ongoing work focuses on assimilating these observations into reanalysis models to improve our picture of historical weather. The dataset will be available through the BORIS portal at the University of Bern. Thank you — I'm happy to take questions.

QUESTIONS:

where can I find the data?

The dataset is not yet publicly available, but we plan to submit the accompanying data descriptor paper by the end of June. Once published, the data will be accessible through the BORIS portal at the University of Bern. If you are interested in the meantime, feel free to contact me directly.

Hello my name is Carlota Corbella and I am a phd student at the university of Bern. Today I am going to present a dataset that I've been working on for almost the past year, and which unifies daily and subdaily weather observations across Europe before 1900.

As you've been hearing throughout this session, these observations exist, but they're scattered across archives, national services, and old publications. HIST-DAILY hopefully brings them together into a single standardized dataset, building on exactly the kind of rescue work many of you are presenting today.

First quickly, why do we need daily data? Monthly databases already give us a lot and monthly reanalyses are quite accurate. But let me show you what monthly data alone can miss. Here's the temperature of Padua through the 1830s.

Only once, in January 1830, did the monthly mean temperature drop below freezing.

That single cold month stands out. But at monthly resolution, that's all we see: one cold dot.

When we superimpose the daily data, we see a different story, with days that swing well above or below the monthly mean.

Daily departures from the monthly mean can reach up to 8 degrees. And this is just for temperature, for precipitation for example the effect is more dramatic, since a single flood day is nothing like continuous light rain, even if the monthly totals match.

We are not only working on daily data, but also on subdaily. The blue bars on this plot show the daily means, but the subdaily observations in temperature show much more variability (as expected). We can see for instance that January 1830 was not only a cold month, but was defined by two pronounced cold waves, and with temperatures reaching to -16C in Turin.

This kind of within-day structure is completely invisible at monthly resolution, and largely invisible even at daily resolution. Just with monthly means we would never know whether the cold was sustained, episodic, or how the events were separated in time. This distinction matters for understanding the atmospheric dynamics.

So let me show you what's in HIST-DAILY first, and then walk through how we built the database. We collected data from 56 sources, both already published and not yet published. In total, more than 1700 series.

The focus has been on pressure and temperature, But we've also included non-instrumental observations such as wind direction, which are valuable for reconstructing synoptic patterns at the daily scale.

This figure shows how the available network evolves over time. Keep in mind that this is what has been digitized and collected, and many more records exist in archives that haven't yet been rescued. A few features stand out. First, the peak in the 1780s is from the Palatine Society, a coordinated European observing network with standardized instruments and one of the earliest attempts at systematic weather monitoring across the continent. Then a long gap through much of the early 19th century, when observations were sparser and more scattered. From the mid-19th century onward, national meteorological services began to be established, and the network grows steadily.

temperature and pressure offer the broadest coverage, from the Iberian Peninsula to Scandinavia and into Russia., Precipitation clusters in the British Isles and the Alpine area, and wind direction is concentrated in central Europe. Coverage is thinner in southeastern Europe and across much of Scandinavia before the mid-19th century.

FLOWCHART. So now let's go to how the data are presented. We collected data from a variety of sources and did an initial and rigorous preprocessing, standardizing units, times and calendar days to the Gregorian calendar which that is of course very important for daily data. Then this pre-processed data was submitted to a quality control where mostly detected digitization errors, daily repetitions, and wrongly written dates. We corrected everything that was very clearly a digitization error by going to the original files but otherwise all other errors were flagged and not corrected.

SEF

Now let's go for a validation example, the cold winter of 1829/30, one of the coldest winters on record in Europe. The top panel shows daily temperature anomalies through this winter for five northern Italian stations in our dataset. The colored lines are the observations and the black line with the gray shading is the 20th century reanalysis ensemble mean, for the corresponding cell where each station is located, and spread. We can see that the anomalies of the 5 independent stations agree remarkably well with each other, which gives us confidence that our quality control preserves real climate signals, and also that the reanalysis follows the observations closely in timing, but often underestimates the depth of the cold extremes and overestimates the warm ones.

We can see this better quantitatively in the bottom left panel and in the middle, where the observed distribution has heavier cold tails than 20CR reproduces.

And on the bottom right we see pairwise correlations between the stations, all above 0.8, which shows internal consistency across independent records.

So with this very brief case study we can see that HIST-DAILY puts together information that current reanalyses still miss.

OUTLOOK